

Data Analyst Evaluation Rubric

SBR Datathon 2026 — HR AI Agent Version 1.0

1. Overview

Purpose

This rubric defines how the HR AI Agent evaluates candidates applying for **Data Analyst** roles. It is the single source of truth for scoring, evidence collection, and final recommendation logic. All prompts, scoring code, and synthetic data generators must reference this document.

Design Principles

- 1. **Generalizable, not memorized.** The agent must evaluate based on *underlying competencies*, not surface keywords. A candidate who says "Structured Query Language" must be evaluated the same as one who says "SQL".
- 2. **Evidence-based.** Every score must be backed by concrete evidence from CV, JD match, or interview answers.
- 3. **Fair under uncertainty.** When evidence is missing, the rubric assigns a *confidence* flag rather than guessing.
- 4. **Multi-source triangulation.** Skill claims in a CV must be validated against interview behavior before they count fully.

Role Definition (from SBR JD)

Collects, cleans, analyzes, and visualizes data to generate insights and support decision-making. **Key responsibilities:** Extract and clean data; write SQL queries; build dashboards; analyze trends; present insights to business stakeholders; validate data quality. **Required skills:** SQL, Excel, Power BI/Tableau, Data cleaning, Statistics basics, Dashboarding, Insight communication.

2. The Five Evaluation Dimensions

#	Dimension	Weight	Maps to SBR Output
D1	Technical Skills & Tools	30%	Knowledge & Skills Score
D2	Analytical & Statistical Thinking	25%	Knowledge & Skills Score

#	Dimension	Weight	Maps to SBR Output
D3	Business Acumen & Insight Communication	20%	Communication Score
D4	Experience & Project Ownership	15%	Fit Score
D5	Role Fit & Soft Signals	10%	Communication Score + Risk Flags

Why these weights? Technical skills (D1) get the largest weight because they are the *non-negotiable* core of the role. Analytical thinking (D2) is the differentiator between a query-runner and a real analyst. Business communication (D3) gets significant weight because dashboards without insight are worthless. Experience (D4) is a proxy — quality matters more than years. Soft signals (D5) catch red flags but rarely make or break a strong candidate.

D1 — Technical Skills & Tools (Weight: 30%)

What it measures: Hands-on proficiency with the core technical tools and methods of data analysis.

Sub-components:

- **SQL proficiency** (querying, joins, aggregation, window functions, optimization)
- **BI/Visualization** (Power BI, Tableau, dashboard design principles)
- **Spreadsheet mastery** (Excel: pivot tables, lookups, advanced formulas)
- **Data wrangling** (Python/Pandas, R, or advanced Excel for cleaning)
- **Database fundamentals** (schemas, joins, normalization basics)

Level Descriptors:

Level	Score	Descriptor	Evidence Indicators
5 — Expert	90-100	Designs queries with window functions, CTEs, query optimization. Builds production dashboards. Comfortable with multiple BI tools. Handles complex data wrangling in Python or advanced Excel.	Mentions specific advanced techniques (window functions, CTEs, performance tuning). Describes optimizing slow queries. Builds dashboards used by ≥10 stakeholders.
4 — Strong	75-89	Writes complex queries with multiple joins, subqueries, aggregations. Builds interactive dashboards. Handles dirty data confidently.	Describes 3+ types of joins correctly. Names dashboard tools used in real projects. Explains specific cleaning steps applied.
3 — Adequate	55-74	Comfortable with basic-to-intermediate SQL (SELECT, JOIN, GROUP BY). Can build simple dashboards from clean data. Uses Excel competently.	Can describe what a JOIN does. Has built ≥1 dashboard. Uses pivot tables. May struggle with edge cases.
2 — Below Bar	30-54	SQL exposure is shallow — can read but not confidently write. Dashboard work is limited to following tutorials. Heavy reliance on others for technical work.	Vague about tools used. Cannot explain a JOIN clearly. No independent dashboard ownership.
1 — Insufficient	0-29	No meaningful exposure to SQL or BI tools. Only academic/coursework exposure.	"Basic communication" listed as a skill. No tool names in experience. Recent coursework only.

What to ask in interview to probe D1:

- "Walk me through how you'd extract monthly active users from a **users** and **events** table."
- "Tell me about a dashboard you built — what decisions did it inform?"
- "How do you handle null values when joining tables?"

Red flags:

- Lists 10+ tools but can't describe a specific use case for any
- Claims advanced skills but project descriptions are generic ("worked on data")
- Tool list contradicts described work (claims Tableau but all projects use Excel)

D2 — Analytical & Statistical Thinking (Weight: 25%)

What it measures: The ability to *reason* with data, not just execute queries. This is the differentiator between a report-builder and a true analyst.

Sub-components:

- **Statistical foundations** (distributions, central tendency, variance, significance)
- **Hypothesis-driven analysis** (forms testable questions before pulling data)
- **Pattern recognition** (identifies trends, anomalies, root causes)
- **Pitfall avoidance** (correlation $\neq$ causation, Simpson's paradox, sampling bias, survivorship bias)
- **Experimentation basics** (A/B test design, sample size intuition)

Level Descriptors:

Level	Score	Descriptor	Evidence Indicators
5 — Expert	90–100	Frames problems as testable hypotheses. Anticipates statistical pitfalls. Discusses confidence intervals, significance, segmentation effects naturally. Designs experiments.	Mentions A/B test design, confounders, Simpson's paradox, or selection bias unprompted. Describes a time they found a counterintuitive insight.
4 — Strong	75–89	Distinguishes correlation from causation. Segments data to find drivers. Validates findings with second data source or sanity check.	Describes investigating root cause behind a metric drop. Mentions validating data before reporting. Talks about distributions, not just averages.
3 — Adequate	55–74	Understands averages, percentages, basic trends. Can identify obvious patterns. May miss subtle pitfalls.	Comfortable describing month-over-month changes. Mentions outliers when asked. Aware that correlation $\neq$ causation when prompted.
2 — Below Bar	30–54	Treats data extraction as the goal. Limited interpretation. Confuses metrics or misapplies statistics.	Describes outputs (charts, reports) but not insights. Uses averages where medians would be appropriate without justification.
1 — Insufficient	0–29	No analytical reasoning evident. Treats analysis as button-pushing.	Cannot describe what makes an insight "actionable." No examples of finding something non-obvious.

What to ask in interview to probe D2:

- "If revenue dropped 15% last month, how would you investigate?"
- "What's the difference between correlation and causation? Give an example from your work."
- "When would you use a median instead of a mean?"

Red flags:

- "Statistics basics" listed without any example of applying statistics
 - Describes analysis projects only in terms of tools used, never findings
 - Cannot articulate what made a past insight valuable
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D3 — Business Acumen & Insight Communication (Weight: 20%)

What it measures: Can the candidate turn data into business value, and explain it to non-technical stakeholders?

Sub-components:

- **Business context** (understands the why behind the metric)
- **Audience adaptation** (adjusts depth for executive vs. peer vs. technical)
- **Storytelling** (structures findings: context → analysis → insight → action)
- **Recommendation quality** (suggests next steps, not just findings)
- **Stakeholder collaboration** (translates business questions to data questions)

Level Descriptors:

Level	Score	Descriptor	Evidence Indicators
5 — Expert	90–100	Naturally frames work in business impact. Tailors communication seamlessly. Drives decisions through analysis.	Quantifies business outcomes (revenue impact, cost savings, decisions influenced). Has presented to executives. Describes pushback and how they navigated it.
4 — Strong	75–89	Connects analyses to business goals. Communicates findings clearly. Recommends actions.	Resume bullets include "led to decision X" or "informed Y%". Can explain a complex analysis simply.
3 — Adequate	55–74	Communicates findings adequately. Sometimes connects to business impact when prompted.	Describes presentations made. Mentions stakeholders worked with. Doesn't always link analysis to action.
2 — Below Bar	30–54	Communicates as a technician — what was done, not why it mattered. Stakeholder interaction is one-way.	Describes deliverables, not impact. Uses jargon when explaining to non-technical audiences.
1 — Insufficient	0–29	No evidence of business communication. Treats reports as the deliverable.	No mention of stakeholders, decisions, or business impact in CV or interview.

What to ask in interview to probe D3:

- "Explain the most complex analysis you've done to someone with no technical background."
- "Describe a time your analysis changed a business decision."
- "How do you handle a stakeholder who wants a dashboard but can't tell you exactly what they need?"

Red flags:

- All project descriptions are technical with zero business context
- Uses heavy jargon when asked to simplify
- Treats analyst as an order-taker role

D4 — Experience & Project Ownership (Weight: 15%)

What it measures: Track record of doing the work end-to-end. Years matter less than depth and ownership.

Sub-components:

- **Years of relevant experience** (calibrated, not absolute)
- **Project complexity** (data volume, stakeholder count, technical difficulty)
- **Ownership level** (executed vs. led vs. owned vs. designed)
- **Measurable outcomes** (quantified impact)
- **Project diversity** (different problem types, domains, data sources)

Level Descriptors:

Level	Score	Descriptor	Evidence Indicators
5 — Expert	90–100	6+ years OR 4+ years with exceptional ownership. Has led end-to-end projects with measurable, large-scale impact.	"Owned X from scoping to delivery." Quantified outcomes ≥ 3 times. Multiple project types.
4 — Strong	75–89	4–6 years with clear ownership in 2+ projects. Quantified outcomes.	"Independently delivered..." Specific metrics on outcomes. Domain variety.
3 — Adequate	55–74	2–4 years OR strong recent ownership in 1+ project.	"Part of a team that..." Some quantification. Limited project variety.
2 — Below Bar	30–54	<2 years OR 2+ years but mostly executing others' work.	"Contributed to..." Few or no quantified outcomes. Generic task descriptions.
1 — Insufficient	0–29	Coursework/internship only. No production ownership.	"Has basic exposure through coursework..." No real-world project bullets.

Calibration notes:

- **Junior role** (0–2 years experience): cap D4 at Level 4 if exceptional, otherwise scale to potential.
- **Senior role** (5+ years): expect Level 4 minimum; Level 3 with 8+ years is a red flag.
- **Career switcher**: rate based on transferable analytical projects, even if not titled "analyst."

Red flags:

- 5+ years but all bullets describe junior-level tasks
 - Job hopping (3+ companies in 3 years) without progression in scope
 - Inflated language ("led", "owned", "designed") without specifics
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D5 — Role Fit & Soft Signals (Weight: 10%)

What it measures: Will this person thrive in *this* role and *this* environment? Catches red flags and validates claims through behavior.

Sub-components:

- **Communication style** (clarity, organization, conciseness — separate from D3 content)
- **Curiosity** (asks questions, learns continuously)
- **Collaboration signals** (works well across functions)
- **Self-awareness** (acknowledges weaknesses honestly)
- **Consistency check** (do interview answers match CV claims?)

Level Descriptors:

Level	Score	Descriptor	Evidence Indicators
5 — Expert	90-100	Crystal-clear communication. Visibly curious. Strong self-awareness. Claims fully validated by interview.	Asks insightful questions. Acknowledges trade-offs in past decisions. CV claims line up perfectly with answers.
4 — Strong	75-89	Clear and organized. Demonstrates learning mindset. Claims largely consistent with behavior.	Structured answers (e.g., STAR format). Mentions learning new tools recently.
3 — Adequate	55-74	Adequate communication. Some signs of growth mindset. Minor inconsistencies between claims and answers.	Answers are correct but unstructured. No major red flags.
2 — Below Bar	30-54	Disorganized or vague communication. CV claims partially contradict answers.	Long, rambling answers. Says "I used Tableau" but can't describe a specific dashboard.
1 — Insufficient	0-29	Cannot communicate clearly OR major inconsistencies between CV and interview.	Off-topic answers. Claims a skill they cannot demonstrate at all.

Red flag triggers (auto-flag for follow-up regardless of score):

- Major contradiction between CV claim and interview answer
- Off-topic responses to direct technical questions
- Inability to discuss any project in specific detail
- Inflated titles inconsistent with described responsibilities

3. Scoring Methodology

Step 1: Level → Numeric Score

Each dimension is evaluated as Level 1-5, then converted:

Level	Numeric Score
Level 5	95
Level 4	82
Level 3	65
Level 2	42
Level 1	20

Why these numbers? They reflect realistic score ranges seen in practice (avoiding artificial flat values like 20/40/60/80/100, which over-cluster). They also align with the distributions seen in the SBR training data.

Step 2: Weighted Aggregation

$$\text{Fit Score} = (D1 \times 0.30) + (D2 \times 0.25) + (D3 \times 0.20) + (D4 \times 0.15) + (D5 \times 0.10)$$

Step 3: Confidence Adjustment

For each dimension, the agent assigns a **confidence level** based on evidence quality:

Confidence	When to assign	Score adjustment
High	Multiple evidence sources agree (CV + interview + tool mentions)	No adjustment
Medium	Some evidence, but gaps remain	No adjustment, but flag in Risks
Low	Single source or contradictions present	Cap dimension score at Level 3 (65), flag in Risks

Rule: Never assign Level 5 with Low confidence. If evidence is thin, the candidate is *Level 3 with low confidence*, not Level 5.

4. Mapping to SBR Required Outputs

The 6 outputs SBR requires are derived from the rubric as follows:

Output 1: Fit Score (0-100)

Fit Score = weighted sum of all 5 dimensions (see Section 3, Step 2)

Output 2: Knowledge & Skills Score (0-100)

Knowledge & Skills Score = $(D1 \times 0.60) + (D2 \times 0.40)$

Pure technical + analytical capability, isolated from communication and experience.

Output 3: Communication Score (0-100)

Communication Score = $(D3 \times 0.70) + (D5_communication_subcomponent \times 0.30)$

D3 captures business communication; D5 captures raw communication style and consistency.

Output 4: Strengths & Weaknesses

Generated by rule:

- **Strength:** Any dimension at Level 4 or 5 with High confidence → add to strengths with evidence
- **Weakness:** Any dimension at Level 2 or below → add to weaknesses with evidence
- **Always include:** at least 2 strengths and 2 weaknesses; if fewer, note explicitly

Output 5: Risks & Uncertainty

Auto-populated when ANY of these triggers fire:

- Any dimension confidence = Low
- Standard deviation across dimensions > 20 points (uneven profile)
- Any red flag triggered (see D1-D5 red flag sections)
- Years of experience inconsistent with role complexity claimed
- Major CV-interview contradiction

Output 6: Final Recommendation

Based on Fit Score thresholds, with risk overrides:

Fit Score	Default Recommendation
≥ 80	Recommend
65-79	Maybe / Needs further assessment
50-64	Conditional / Requires follow-up interview
< 50	Do not recommend

Override rules:

- If **any** red flag is triggered → minimum recommendation = "Conditional / Requires follow-up interview" (even if Fit ≥ 80)
- If **D1 < Level 3** (technically insufficient) → cap recommendation at "Maybe" regardless of Fit Score
- If **standard deviation across dimensions > 20** → recommend follow-up interview to verify outlier dimension

5. Evidence Collection Guide

Where to look for each dimension:

Dimension	CV signals	Interview signals
D1 Technical	Tools list, specific techniques in bullets, certifications	"Walk me through..." answers, ability to describe a JOIN, tool-specific stories
D2 Analytical	Project descriptions with insights (not just outputs), statistical methods mentioned	Hypothesis-framing questions, "How would you investigate..." answers
D3 Business	Quantified business outcomes, stakeholder mentions, presentation experience	"Explain to non-technical audience..." answers, decision-impact stories
D4 Experience	Years, titles, project descriptions, ownership verbs	Specificity of project details when probed
D5 Soft signals	Resume structure and clarity	Consistency check vs. CV, communication style during interview

Handling Missing Evidence

- **Never** infer missing evidence as positive.
 - If a dimension has no supporting evidence after probing: assign Level 2 (cap) with Low confidence and flag in Risks.
 - If interview answers are missing entirely: max possible Fit Score = 70, with "Requires interview completion" flag.
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6. Calibration Rules — Anti-Memorization

The rubric must generalize, not memorize. These rules protect against overfitting:

Rule 1: Skill validation, not skill listing

A candidate who lists "SQL" gets D1 baseline credit only. To move above Level 3 in D1, they must *demonstrate* SQL knowledge in interview (describe a JOIN, explain a query, troubleshoot an error).

Rule 2: Synonym-blind matching

The agent must treat semantic equivalents as identical:

- "SQL" = "Structured Query Language" = "T-SQL" = "PL/SQL"
- "Power BI" = "PowerBI" = "MS Power BI"
- "Data cleaning" = "Data wrangling" = "Data preparation"
- "BI" = "Business Intelligence"
- "Stats" = "Statistics" = "Statistical analysis"

Rule 3: Don't double-count tool listings

If a CV lists "SQL, SQL, SQL" or repeats tools in Core Skills and Tools sections, count once.

Rule 4: Penalize vague claims

"Basic communication" as a listed skill → automatic D5 ceiling at Level 3 (it's a filler, not a skill).

"Has basic exposure to..." → D4 ceiling at Level 2.

Rule 5: Reward specificity

Quantified outcomes ("reduced query time by 40%", "dashboard used by 30 stakeholders") always score higher than vague ones ("improved performance", "shared with team").

Rule 6: Context-aware weighting for unusual profiles

Career switchers, bootcamp graduates, and gap candidates may have non-traditional CVs. The agent should:

- Lower D4 weight to 10% (from 15%)
 - Raise D5 weight to 15% (consistency check is more important)
 - Add follow-up question quota: 2 additional probing questions on transferable skills
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7. Example Score Computation

Candidate: CV-107 (Sara Alfaisal, Strong tier)

Evidence summary:

- 5 years, BSc Information Systems, CFA Level I
- Skills: SQL, Excel, Power BI/Tableau, Data cleaning, Statistics basics, BigQuery, Tableau, Power BI
- Experience: "Cleaned datasets in 3+ projects with measurable impact"
- Interview answers (synthetic, for illustration): demonstrates JOIN knowledge clearly, describes dashboard for 15 stakeholders, mentions A/B test validation

Dimension scoring:

Dimension	Level	Score	Confidence	Justification
D1 Technical	4	82	High	SQL + BI tools demonstrated; BigQuery is bonus
D2 Analytical	4	82	High	Mentions A/B test validation; describes investigation flow
D3 Business	4	82	Medium	Quantified stakeholders; needs more on decision impact
D4 Experience	4	82	High	5 years, 3+ projects with measurable impact
D5 Soft signals	5	95	High	Clear, consistent, structured answers

$$\begin{aligned}\text{Fit Score} &= (82 \times 0.30) + (82 \times 0.25) + (82 \times 0.20) + (82 \times 0.15) + (95 \times 0.10) \\ &= 24.6 + 20.5 + 16.4 + 12.3 + 9.5 \\ &= 83.3 \rightarrow \text{Recommend}\end{aligned}$$

Knowledge & Skills Score = $(82 \times 0.60) + (82 \times 0.40) = 82$ Communication Score = $(82 \times 0.70) + (95 \times 0.30) = 85.9$

Strengths: Strong technical foundation; clear communicator; consistent profile. **Weaknesses:** Business impact storytelling could be deepened. **Risks:** None major; one Medium-confidence dimension noted. **Recommendation:** Recommend.

✓ This matches SBR's labeled "Expected Fit Score: 95.0, Recommend" within reasonable range (we're slightly more conservative on D3, which is realistic without seeing actual interview).

8. Open Questions for Calibration (Pre-Implementation)

Before coding the agent, the team should align on:

1. **Junior vs. Senior calibration:** Does this rubric serve all levels, or do we build separate weight profiles?
 2. **Industry adjustments:** Should finance-domain analysts be weighted differently than e-commerce?
 3. **Language handling:** If interview is in Arabic vs. English, does the rubric need adjustments for D3/D5?
 4. **Threshold validation:** Run rubric against all 35 labeled Data Analyst CVs and compare predicted vs. expected scores. Adjust weights if systematic bias appears.
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End of Rubric v1.0 *This document is the foundation for: synthetic data generation, prompt design, scoring code, and evaluation metrics.*